

Introduction to Polynomial Sequences: A Survey of Fibonomials and their Applications

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5 ABSTRACT: Polynomial sequences are incredibly prevalent in the field of mathematics.
6 Oftentimes when working with a set of numbers, turning them into a polynomial sequence
7 introduces new methods of approaching problems and new properties that can be utilized to find
8 a solution. One such polynomial sequence that has been widely utilized is the Fibonomials, a
9 sequence stemming from the well-known Fibonacci numbers. This polynomial sequence, and
10 its resulting class of sequences, both further our knowledge of the Fibonacci numbers and are
11 incredibly applicable to other fields. Through a study of the Fibonomials, this survey paper aims
12 to introduce the reader to the uses of polynomial sequences and encourage them to apply such
13 sequences to their own work. We will introduce the Fibonomials and their related polynomials,
14 discuss the properties of the Fibonomials and their uses, demonstrate some of the problems that
15 have been approached using Fibonomials, and make connections between sequences in order to
16 highlight the broader use of polynomial sequences. Ultimately, we hope to inspire future work
17 utilizing polynomial sequences.

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1. Introduction

In this paper, we will primarily be studying the Fibonomials, a specific polynomial sequence. Stemming from the Fibonacci numbers, this polynomial sequence has been studied on and off throughout the years. First introduced by François Lucas, this polynomial sequence has since been utilized by a wide variety of mathematicians.

François Lucas was a French mathematician primarily focused on number theory. His interest in polynomial sequences stemmed from his work in primality testing. He constructed his own numerical sequence that could be used to this end, called the Lucas sequence. Lucas then realized that the well known Fibonacci numbers were a special case of his sequence, leading him to his work with the Fibonomials.

In general, a polynomial sequence is understood as an indexed collection of polynomials; the useful ones have some recurrence or other relationship to each other or some object of interest. Such polynomials can stem from a specific numerical sequence, a method of polynomial generation, or some other calculation. We can see from this broad definition that polynomial sequences are applicable to a large swath of mathematical work.

Whenever numerical sequences and recurrences appear in mathematicians' work, polynomial sequences can easily be utilized. Due to their numerous properties, which are unique to each sequence and their origins, they provide many insights into mathematical problems. Moreover, polynomial sequences often relate to each other and are studied in very similar ways, leading to vast amounts of literature surrounding specific sequences. In this way, researchers in need of a novel approach to their problem can gain immense benefit from the utilization of polynomial sequences.

This paper sets out to thoroughly describe the Fibonomials and their uses and connections to other sequences. We hope that from this researchers are able to gain inspiration on how to incorporate polynomial sequences in their work.

2. Preliminaries

a. Fibonacci Numbers and their Polynomials

We begin with a definition of the Fibonomials. The Fibonacci numbers have been well-known and well-studied for many centuries. These numbers arise from the recurrence

$$F_n = F_{n-1} + F_{n-2}, \quad n \geq 2$$

for $n \geq 2$, with F_0 and F_1 defined to be 0 and 1 respectively. The first step to consider related polynomial sequences is to modify this definition to generate polynomials, as opposed to constants. To that end, we define the resulting Fibonacci Polynomial as such:

Definition 1 *The n th Fibonacci polynomial is the polynomial $F_n(x)$ which satisfies the recurrence*

$$F_n(x) = xF_{n-1}(x) + F_{n-2}(x)$$

with initial conditions $F_0(x) = 0, F_1(x) = 1$.

It is clearly that each polynomial $F_n(x)$ roughly corresponds to the n th Fibonacci number (Benjamin and Quinn (2003)). We can clearly see from the recurrence that taking $x = 1$ just gives the Fibonacci numbers. Researchers then went a step further, considering generalizations to a polynomial in two variables. The general definition of the Fibonomials then arises.¹

Definition 2 *The n th Fibonomial is the polynomial $f_n(s, t)$ which satisfies the recurrence*

$$f_n(s, t) = sf_{n-1}(s, t) + tf_{n-2}(s, t)$$

with initial conditions $f_0(s, t) = 0, f_1(s, t) = 1$ (Ambdeberhan et al. (2014)).

This sequence was first considered by Hoggatt and Long (Hoggatt and Long (1974)). Note that this definition creates a polynomial sequence. The previously mentioned Fibonacci Polynomial is the special case where $t = 1$. For completeness, we will list out the first few Fibonomials:

$$f_0(s, t) = 0$$

¹The literature on this subject is not very unified: some call these 2-variable polynomials the Lucanomics and denote them $\{n\}$ or $\{n\}_{s,t}$ instead of $f_n(s, t)$

$$f_1(s, t) = 1$$

$$f_2(s, t) = s$$

$$f_3(s, t) = s^2 + t$$

$$f_4(s, t) = s^3 + 2st$$

b. Generating Functions

Another common way to express these polynomial sequences is through generating functions. Given a sequence a_n , its generating function is defined as

$$G(z) := \sum_n a_n z^n$$

It follows that we can then write the generating function of the Fibonacci numbers:

$$\frac{1}{1 - z - z^2}$$

Note that the Fibonacci numbers are not achieved by changing the value of z , but rather the Taylor expansion of this rational function has the Fibonacci numbers as coefficients.

The generating function of the Fibonomials is quite similar to that of the Fibonacci numbers:

$$\sum_{n \geq 0} f_n(s, t) z^n = \frac{z}{1 - sz - tz^2}$$

Note that this function simply re-indexes the polynomials and then adds the variables of s, t . From this, we can note that the generating function of a polynomial sequence shares many similarities to that of its origin. Generating functions express polynomial sequences neatly and more obviously, which allows for connections to be seen more easily. These methods also generalize neatly to studying sequences of polynomials, given a sequence $P_n(x)$, its generating function is

$$G(x, z) = \sum_n P_n(x) z^n$$

c. Gibonacci Numbers and their Polynomials

The natural next step is to explore the other ways in which these polynomials can be manipulated. Up until this point the initial values of the Fibonacci numbers, polynomial, and Fibonomial have remained constant. However, changing these results in what is known as the Gibonacci numbers - or generalized Fibonacci numbers.

Definition 3 *A Gibonacci sequence is any sequence G_n that satisfies the Fibonacci recurrence:*

$$G_n = G_{n-1} + G_{n-2}, \quad n \geq 2$$

Note that the initial conditions are not restricted to being particular constants. From this, the definition of the Gibonacci polynomial $G_n(x)$ and the gibonomial $g_n(s, t)$ are simple to generalize.

Definition 4 *A Gibonacci polynomial sequence is any polynomial sequence in one variable $G_n(x)$ which satisfies*

$$G_n(x) = xG_{n-1}(x) + G_{n-2}(x)$$

Definition 5 *A Gibonomial sequence is any polynomial sequence in two variables $g_n(s, t)$ which satisfies*

$$g_n(s, t) = sg_{n-1}(s, t) + tg_{n-2}(s, t)$$

We can see that such a definition generates a larger class of polynomial sequences. This family of sequences ends up having its own set of useful and related properties.

One last useful and related polynomial sequence is the Lucas Polynomial or Lucanomials. These are named after Lucas, the aforementioned first mathematician to define the Fibonomials. They are a specific case of Gibonacci numbers with initial conditions $L_0 = 2, L_1 = 1$. The Lucas Polynomial $L_n(x)$ and Lucanomial $l_n(s, t)$ initial conditions easily follow: $L_0(x) = 2, L_1(x) = x, l_0(s, t) = 2, l_1(s, t) = s$

d. Summary of Notation

Below is a table summarizing the sequences and notational conventions described previously. Particular attention should be paid to the first three entries, as those come up more frequently.

F_n	The n th Fibonacci number, starting at $F_0 = 0, F_1 = 1$.
$F_n(x)$	The n th Fibonacci polynomial, defined by the recurrence $F_n(x) = xF_{n-1}(x) + F_{n-2}(x)$ for $n \geq 2$, with $F_0(x) = 0, F_1(x) = 1$
$f_n(s, t)$	The n th Fibonomial, defined by the recurrence $f_n(s, t) = sf_{n-1}(s, t) + tf_{n-2}(s, t)$, with $f_0(s, t) = 0, f_1(s, t) = 1$.
G_n	Any Gibonacci sequence, that is, any sequence which obeys $G_n = G_{n-1} + G_{n-2}$ for $n \geq 2$.
$G_n(x)$	Any sequence of Gibonacci polynomials, satisfying $G_n(x) = xG_{n-1}(x) + G_{n-2}(x)$ for $n \geq 2$
$g_n(s, t)$	Any sequence of Gibonomials, which satisfy $g_n(s, t) = sg_{n-1}(s, t) + tg_{n-2}(s, t)$ for $n \geq 2$
L_n	The Lucas numbers, a specific Gibonacci sequence with $L_0 = 2, L_1 = 1$.
$L_n(x)$	The Lucas Polynomials, a specific Gibonacci polynomial sequence with $L_0(x) = 2, L_1(x) = x$
$l_n(s, t)$	The Lucanomials, a specific Gibonomial sequence with $l_0(s, t) = 2, l_1(s, t) = s$.

3. Properties of the Fibonomial

We now move on to the specific properties of these polynomial sequences. Understanding and finding such properties is incredibly useful in the study of polynomial sequences, as they both provide additional tools to solve our overall mathematical problem and relate back to the origins of the polynomial sequence. Interesting properties of the polynomial sequence often indicate analogous properties of the sequence's origin. In this case, the origin of our polynomial sequence is the Fibonacci numbers.

There are a multitude of identities for polynomial sequences, just as there are for the Fibonacci and Lucas numbers. A selection is presented below.

$$f_n(2, -1) = n$$

This property comes trivially by induction, as $2(n-1) - (n-2) = n$ (Ambdeberhan et al. (2014)). This relationship to the natural numbers allows the Fibonomials to be applied to induction.

$$f_n(s, t) = t^{\frac{n-1}{2}} F_n\left(\frac{s}{\sqrt{t}}\right)$$

This property gives a nontrivial connection between Fibonomials and Fibonacci polynomials. These sorts of identities are useful because it means that we can study polynomial sequences by studying related sequences, and converting properties through these formulas (Ambdeberhan et al. (2014)). Furthermore, it allows results obtained on one polynomial sequence to apply to the other.

$$f_{m+n}(s, t) = \frac{l_m(s, t)f_n(s, t) + f_m(s, t)l_n(s, t)}{2}$$

This property shows the relation between the Fibonomials and Lucanomials well. It also indicates that $l_n(s, t) | f_{2n}(s, t)$.

$$l_n(s, t) = f_{n+1}(s, t) + t f_{n-1}(s, t)$$

It is a well-known result that the Lucas numbers are the sum of sandwiching Fibonacci numbers: that is $L_n = F_{n+1} + F_{n-1}$. This property of the Fibonomials and Lucanomials is a nice extension of that. (Ambdeberhan et al. (2014)). Just as the Fibonacci numbers and Lucas numbers are closely related, so too are the Fibonomials and Lucanomials.

$$f_{n+m}(s, t) = f_m(s, t)f_{n+1}(s, t) + t f_{m-1}(s, t)f_n(s, t)$$

$$\gcd(f_m(s, t), f_n(s, t)) = f_{\gcd(m, n)}(s, t)$$

This first property is a nice alternative to the previous one in lieu of the Lucanomials. These properties are both particularly elegant and are examples of intrinsically interesting properties of polynomial sequences that might be useful when considering problems the sequences originate from (Ambdeberhan et al. (2014)). It is also non-trivial, which demonstrates how these sequences can be meaningfully extended to number theory. Calculating the greatest common divisor of two numbers is an important mathematical problem, and thus solutions such as these allow us to make advances.

$$f_n(s, t) = \sum_{k \geq 0} \binom{n-k-1}{k} s^{n-2k-1} t^k$$

These sorts of explicit formulas for polynomial sequences are often useful in finding other identities (Ambdeberhan et al. (2014)). We can also work directly from this closed form expression, which can lead to further properties and identities regarding the Fibonomials.

$$\sum_{k=0}^n \binom{n}{k} f_k(s, t) = f_n(s+2, t-s-1)$$

This gives an interesting relationship between each Fibonomial and its predecessors (Ambdeberhan et al. (2014)). It also further establishes the connection between polynomial sequences, allowing us to work with many of the aforementioned sequences.

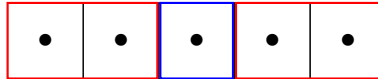
4. Connecting Fibonomials to Other Fields

a. Combinatorics: Tilings of Dominoes and Monominoes

One common interpretation of the Fibonacci numbers is counting the number of tilings of a $1 \times n$ rectangle with dominoes and monominoes.

Theorem 1 *A $1 \times n$ rectangle can be tiled using monominoes and dominoes in F_n different ways (Ambdeberhan et al. (2014)).*

The Fibonomials provide us a way to expand this problem by assigning weights to the tilings. Assign a weight s to each monomino in the tiling and weight t to each domino in the tiling. We then define the overall weight of the tiling to be the product of each monomino/domino in the tiling. For example, view the following picture where each red rectangle is a 1×2 domino and the blue square represents a 1×1 monomino.



The weight of this tiling would be $s \cdot t^2$ as there is one monomino and two dominoes. In general, the weight of each tiling will be $s^{(\# \text{ of monominoes})} \cdot t^{(\# \text{ of dominoes})}$ (Ambdeberhan et al. (2014)).

Theorem 2 *The sum of the weights of the tilings of a $1 \times n$ rectangle as described above is exactly the $n+1$ th Fibonomial $f_{n+1}(s, t)$*

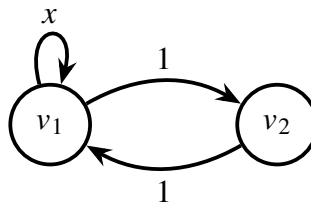
This draws a parallel between combinatorial interpretations of Fibonacci numbers and Fibonomials. Note that the theorem follows from the recurrence inductively: For a base case, consider $n = 0$ and $n = 1$. For $n = 0$, we take the convention that the weight of the empty tiling is $1 = f_1(s, t)$, as the exponents of s, t are both 0. For $n = 1$, clearly, a 1×1 rectangle has total weight in tilings of s , as it only has 1 possible tiling, and that tiling is one monomino, and so has weight $s = f_2(s, t)$.

Now consider $1 \times n$ rectangle: the last tile can either be a domino or a monomino. Therefore the weighted sum of a $1 \times n$ rectangle is equal to s times the weighted sum of a $1 \times (n - 1)$ rectangle (by appending a monomino to each tiling of a $1 \times (n - 1)$ rectangle) plus t times the weighted sum of a $1 \times (n - 2)$ rectangle (by appending a domino to each tiling of a $1 \times (n - 2)$ rectangle). This is precisely the recurrence for the Fibonomials. Since this counting follows the same recurrence and has the same (but offset) initial conditions, we can see that the weighted sum of tilings of a $1 \times n$ rectangle is $f_{n+1}(s, t)$ (Ambdeberhan et al. (2014)).

We can see from this example that polynomial sequences allow us to extend the properties of number sequences to a larger class of problems. By varying the weights of the recurrence, we can count these more complex objects. In particular, we can count weighted versions of everything that the original number sequence counts.

b. Graph Theory: Adjacency Matrices and Weighted Walks

The Fibonomials are also useful in areas previously unrelated to the Fibonacci numbers. Consider, for example, the following weighted directed graph,



One common question to ask about graphs is the total number of walks on them. A walk in a graph is a sequence of vertices with the property that two consecutive vertices in the sequence have an edge between them. These walks can be thought of as starting on a vertex and traveling along edges around the graph. In particular, since this is a weighted graph, we assign to each walk a weight as the product of the weights of edges traversed by the walk and are in-

interested now in the sum of the weights of the walks. From graph theory, we get the following result.

Theorem 3 *The sum of the weights of the walks from v_i to v_j of length n on a graph is the i, j entry of A^n , for A the adjacency matrix of the graph (Koshy (2019)).*

The adjacency matrix of a graph has its i, j entry as the weight of the edge from v_i to v_j , or 0 if there is no edge, so the adjacency matrix of the above graph is

$$A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$$

One observation to be made about this matrix is that the first row contains the coefficients of the Fibonacci polynomial recurrence. From this we can see that matrix multiplication yields,

$$\begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} F_n(x) \\ F_{n-1}(x) \end{bmatrix} = F_{n+1}(x)$$

Thus, since we can see that

$$A = \begin{bmatrix} F_2(x) & F_1(x) \\ F_1(x) & F_0(x) \end{bmatrix}$$

we get inductively that

$$A^n = AA^{n-1} = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_n(x) & F_{n-1}(x) \\ F_{n-1}(x) & F_{n-2}(x) \end{bmatrix} = \begin{bmatrix} F_{n+1}(x) & F_n(x) \\ F_n(x) & F_{n-1}(x) \end{bmatrix}$$

so that the sum of the weights of the walks of fixed length and endpoints on this graph is counted by the Fibonacci polynomials (Koshy (2019)). Observing this connection between an understood polynomial sequence and this problem—counting weighted walks by evaluating powers of an adjacency matrix—gives a more broadly applicable idea. Given any such problem of this form, we can convert it into a question about polynomial sequences, and use the tools developed to understand the resulting sequences, and thus the original problem. Additionally, if we find a polynomial sequence with a recurrence that can be expressed as the powers of a matrix, then we

can convert it into a matrix, and thus a graph, and analyze them by applying linear algebra and graph theory tools to the resulting constructions to understand the sequence.

5. Connections Between Polynomial Sequences

a. An Extension of the Fibonomials

We've discussed how the Fibonomials fit into the larger polynomial class of the Gibonomials, but these connections can be expanded even further. The Fibonomials are related to a completely different set of polynomials - the Chebyshev polynomials. There are two kinds of Chebyshev polynomials, the first kind is a set of orthogonal polynomials under the inner product $\int_{-1}^1 \frac{f(x)g(x)}{\sqrt{1-x^2}} dx$. They are defined by generalizing the double angle formulas: $T_n(\cos(\theta)) = \cos(n\theta)$. They can be equivalently defined by the recurrence $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$ with initial conditions $T_0(x) = 1, T_1(x) = x$. They also come from the following formula:

$$U_n(\cos(\theta)) = \frac{\sin((n+1)\theta)}{\sin(\theta)}$$

(Szeg (1939)). Their generating function is given by $\frac{1-tx}{1-2tx+t^2}$. The Chebyshev polynomials of the second kind are very closely related to the first. They are orthogonal under the inner product $\int_{-1}^1 f(x)g(x)\sqrt{1-x^2}dx$. They follow the same recurrence with different initial conditions: $U_0(x) = 1, U_1(x) = 2x$. Their generating function is given by $\frac{1}{1-2xt+t^2}$.

At first glance, these polynomials seem entirely unrelated to the Fibonomials. However, their generating functions are very similar. In fact, they differ exactly by a variable substitution, namely $s = 2x, t = -1$. Thus, we get the following theorem.

Theorem 4

$$U_n(x) = f_n(2x, -1)$$

This means that the Chebyshev polynomials are closely related to the Fibonomials—an unexpected result.

From this, we gain two insights. Firstly, generating functions are an incredibly useful tool for evaluating polynomial sequences. Due to their novel form, they often carry different information and allow us to easily see relationships between sequences. Secondly, we have shown that

polynomial sequences are interconnected. Finding one sequence often allows you to relate it to other better-studied polynomial sequences. These relationships can then be utilized to study the properties of your polynomial sequences, as finding a relationship like this between them allows them to inherit each other's properties.

b. Further Expansion to Pascal's Triangle

These connections expand even further. We can use the same method of inspecting the generating function to find more related polynomial sequences. First note that the Chebyshev polynomials of the second kind are actually special cases of the ultraspherical Gegenbauer polynomials $C_n^{(\alpha)}(x)$. The superscript α here is a parameter: the Gegenbauer polynomials are thought of as polynomials in one variable x . There is a sequence of polynomials for each α . Their generating function is given by $\frac{1}{(1-2xt+t^2)^\alpha}$. From this we can immediately see that the Chebyshev polynomials of the second kind obey $C_n^{(1)}(x) = U_n(x)$ and many more related polynomial sequences arise from changing this α value.

We can then relate the Gegenbauer polynomial to a certain number triangle characterizing twin primes. It is well known that Pascal's triangle can be expressed as the polynomial sequence $(x+1)^k$ where k indexes the rows and the coefficients of the polynomial correspond to the numbers in that row. Another well-known result of Pascal's triangle is that the entries of the k th row of the triangle except the ends are all divisible by k if and only if k is prime. This property was then used to construct a similar triangle characterizing twin primes (Dilcher and Stolarsky (2005)). The formula for the triangle is

$$a(k, s) = \binom{k+s}{2s+1} \frac{(2k+1)(2k-1)}{2s+3}$$

and it turns out that the k th row, obtained by varying $s \in \{0, 1, 2, 3, \dots, k-1\}$ and fixing k , is divisible by $2k+1$ and $2k-1$ with exactly one exception each if and only if $2k+1$ and $2k-1$ were a twin prime pair. This is a significant primality test for twin primes. Since this is a number triangle, it can be used to generate polynomials from its rows, similar to Pascal's triangle. For example, take the $k=3$ row, with entries 35, 28, 5. Indexing these by powers of x yields the row polynomial $35x^2 + 28x + 5$, similar to how the rows of Pascal's triangles are the coefficients of $(1+x)^k$. In general, we define the row polynomials as such:

Definition 6 *The n th row polynomial of the twin prime triangle is*

$$Q_n(x) = \sum_{i=0}^{n-1} a(n, i) x^i$$

This transformation then becomes incredibly helpful in studying the properties of their triangle. In particular, it allows us to use generating functions to study its properties.

The generating function of this polynomial sequence is as follows:

$$\sum_n Q_n(x) t^n = \frac{t(1 + (6+x)t + t^2)}{(1 - (2+x)t + t^2)^2}$$

We notice that this generating function has many similarities to that of the aforementioned Gegenbauer polynomials. Inspection thus yields

$$\sum_n Q_n(x) z^n = t(1 + (6+x)t + t^2) \sum_n C_n^{(2)}\left(\frac{x+2}{2}\right) t^n$$

and standard sum manipulations to collect like powers of t give a formula for their polynomial in terms of Gegenbauer polynomials,

Theorem 5

$$Q_n(x) = C_{n-1}^{(2)}\left(\frac{2+x}{2}\right) + (6+x)C_{n-2}^{(2)}\left(\frac{2+x}{2}\right) + C_{n-3}^{(2)}\left(\frac{2+x}{2}\right)$$

(Dilcher and Stolarsky (2005)).

From this, we see that Fibonomials, Fibonacci polynomials, Gibonomials, Chebyshev polynomials, Gegenbauer polynomials, and the specifically constructed twin prime triangle polynomials are all connected. In this way, it is clear how the use of polynomial sequences and the tools to study them can be useful, as it makes spotting these and similar connections near trivial.

6. Further Polynomial Sequences

a. Catalan Polynomials

The Catalan numbers are another well-studied and useful number sequence that we can generalize to polynomials. We know that

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

In order to turn these into polynomials, we need to define a binomial coefficient that outputs polynomials. For this, we replace integers with Fibonacci polynomials. Defining $f_n!(s, t) = f_n(s, t)f_{n-1}(s, t) \cdots f_1(s, t)$, we get

$$\binom{f_n(s, t)}{f_k(s, t)} = \frac{f_n!(s, t)}{f_k!(s, t)f_{n-k}!(s, t)}$$

You can see that this definition of

$$\binom{f_n(s, t)}{f_k(s, t)}$$

is very similar to the standard definition of $\binom{n}{k}$: indeed it preserves many of the most important properties: including (surprisingly) that it produces an integer coefficient polynomial.

Now we can define the Catalan Polynomials as

$$C_n(s, t) = \frac{1}{f_{n+1}(s, t)} \binom{f_{2n}(s, t)}{f_n(s, t)}$$

Perhaps more surprisingly, these are still integer coefficient polynomials.

Another way to express the Catalan numbers is a well-known identity:

$$C_n = \binom{2n-1}{n-1} - \binom{2n-1}{n-2}$$

Fascinatingly, this identity holds for the Catalan polynomials as well. It turns out that

$$\frac{1}{f_{n+1}(s, t)} \binom{f_{2n}(s, t)}{f_n(s, t)} = \binom{f_{2n-1}}{f_{n-1}} - \binom{f_{2n-1}}{f_{n-2}}$$

naturally for $n > 2$ (Ambdeberhan et al. (2014)).

b. Twin Prime Polynomials

Recall the previously described number triangle that characterizes twin prime pairs. Note that the authors constructed a polynomial sequence from it by taking each row, and multiplying its entries by increasing powers of x . In doing this, they found a polynomial recurrence for their triangle,

$$Q_{n+4}(x) - (4+2x)Q_{n+3}(x) + (6+4x+x^2)Q_{n+2}(x) - (4+2x)Q_{n+1}(x) + Q_n(x) = 0$$

This recurrence was useful, as observing that the coefficients of the recurrence and the initial condition were integers gave them a trivial inductive proof to give that their entire triangle was integers (Dilcher and Stolarsky (2005)). This technique is also generally applicable. Given any formula $f(k, s)$ generating a 2d array of numbers, you can convert it to a polynomial sequence in the same way, and studying the sequence can generate insight into the properties of the entries of the array f .

c. Legendre Polynomials

We previously discussed the Chebyshev and Gegenbauer polynomials but did not place emphasis on their origins. However, we now turn our attention to the Legendre polynomials, a set of Gegenbauer polynomials with $\alpha = \frac{1}{2}$. This connection was not established until much later, as the Legendre Polynomials first arose from applying the Gram-Schmidt process to the set of polynomials $\{1, x, x^2, \dots\}$. The Legendre polynomials are a complete set of single variable orthogonal polynomials under the standard inner product $\int_{-1}^1 f(x)g(x)dx$. They arose from the need for orthogonal polynomials, making them in many respects the simplest orthogonal basis of polynomials. This makes them very useful in Physics, Chemistry, and Linear Algebra. They follow the recurrence $(n+1)P_{n+1}(x) = 2(n+1)xP_n(x) - nP_{n-1}(x)$ with initial conditions $P_0(x) = 1, P_1(x) = x$. More interestingly, they have a generating function $\frac{1}{\sqrt{1-2xt+t^2}}$. The most important thing to note about the Legendre Polynomials is that they arose from a calculation, namely, the Gram-Schmidt process described before.

7. Conclusion and Further Study

In this paper, we have extensively covered the Fibonomials. They are an incredibly interesting polynomial sequence that comes from the Fibonacci numbers and share many properties with them. We have seen how they can be used in combinatorics, linear algebra, and graph theory to revisit and expand known facts. We have also seen how the Fibonomials have far-reaching connections, including the Chebyshev, Gegenbauer, and twin-prime polynomials. Lastly, we discussed the various ways to extract polynomial sequences from mathematical work.

From this, it is hopefully clear that polynomial sequences are a powerful tool for mathematical studies. The methods used in this paper, such as observing generating functions, looking at topics related to Fibonacci numbers, and connecting polynomial sequences, are applicable to any such polynomial sequence. Furthermore when working with any of the discussed polynomial sequences, one can rely on properties of the Fibonomials and their larger class of polynomials.

If you wish to continue your study of polynomial sequences, we recommend you look more deeply into the Chebyshev and Gegenbauer polynomials. We also recommend the super Fibonacci and Catalan polynomials and the Jacobsthal polynomials. These all have interesting properties and applications that can give an insight into the incredibly broad uses of polynomial sequences.

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